Tissues in Action

Class :- 9th

Cell: The smallest unit of life that makes up all living organisms. Cells carry out all basic life functions.

Organisms: All living things are called organisms. They may be unicellular (Amoeba) or multicellular (plants and animals).

Tissue: A group of cells that are similar in structure and function. Different types of tissues together form organs, different organs group together, forming organ systems; these systems further form a complete organism.

- Plants and animals differ in nutrition and tissue functions.
- Animals have tissues for digesting food, while plants have tissues that help them utilise solar energy for photosynthesis to make their own food.
- Both have tissues for transport, but their growth patterns differ due to different types of growth tissues.

Plant Tissues: Specialised tissues that help plants synthesise food through photosynthesis, transport water and nutrients, provide support,and control growth.

Plants grow in different ways-by increasing length (height of stem and depth of roots), increasing girth (thickness of stem), and regrowing after being cut or grazed.

Based on the dividing capacity, plant tissues are of two types: **meristematic tissues** and **permanent tissues.**

Meristematic Tissues

Plants grow in three main ways

- Increase in length: The stem grows taller and roots grow deeper into the soil.
- Increase in girth: The stem becomes thicker over time.
- Regrowth after cutting: Branches grow back after being cut or grazed by animals.
- All of this growth requires cells that divide actively and continuously. These cells together form meristematic tissue.
- Plants have three types of meristematic tissues, each located in a different part of the plant and responsible for a different kind of growth.

❖ **Meristematic Tissues:** Consist of actively dividing cells. Meristematic tissues are of **three types**:

- **Apical Meristem**
  - It is found at the tips of roots and shoots.
  - These tip cells divide continuously, pushing the root deeper into the soil and the shoot higher into the air.
  - When the root tip is cut off, the root stops growing. This confirms that length growth occurs only from the tip. **The apical meristem is responsible for growth in length.**

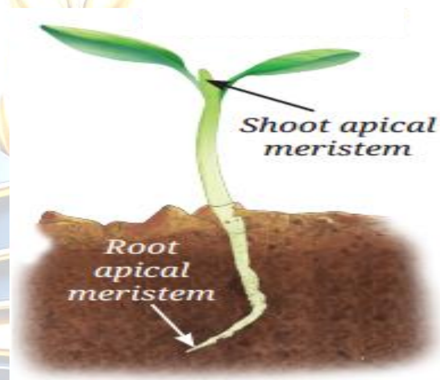


Fig. 3.3: Location of apical meristem in a

- **Lateral Meristem**
  - It runs along the circumference of the stem.
  - Its cells divide and add new cells both inward and outward, causing the stem to become wider over time.
  - A cross-section of a tree trunk shows ring-like patterns. Each ring represents one year of lateral meristem activity and these rings are known as annual growth rings.
  - Wide rings indicate a year with good growing conditions.
  - Narrow rings indicate a year with poor or dry conditions.
  - **By counting these rings**, scientists can estimate the age of a tree and understand past climate conditions. The lateral meristem is responsible for growth in girth (thickness).

Fig. 3.4: T.S. of a tree trunk showing annual growth rings

- **Intercalary Meristem**
  - **It is found at the base of an internode or just above a node.**
  - The point on a stem where branches or leaves arise is called **a node.**
  - The part of the stem **between two nodes is known as internode.**
  - When a stem tip is cut, new branches grow out from the nodes due to this meristem. This gives trimmed hedges a bushy appearance.

- **Grass grows back quickly after being mowed because of intercalary meristems at the nodes of the grass stem.**
- The intercalary meristem is responsible for regrowth after cutting or grazing.

Differentiation: From Meristematic to Permanent Tissue

- Meristematic tissue keeps adding new cells through continuous division.
- Some newly formed cells remain meristematic and keep dividing.
- Others lose the ability to divide, undergo structural changes, and become specialised for specific functions such as support, transport, or storage.
- This process of specialisation is called differentiation. The cells that have differentiated and can no longer divide are called permanent tissues.

Permanent Tissues:- the cells in the tissue that loose the ability to divide

- Permanent tissues are divided into two main categories
- **Simple permanent tissues**:
  These are made of only one type of cell. All cells have the same structure and function.
- **Complex permanent tissues**:
  These are made up of more than one type of cell, all working together for a common function.

✓ **Differences between Parenchyma, Collenchyma and Sclerenchyma:**
✓ **Conducting Tissues-Complex permanent tissue: Made up of more than one type of cells.**

Permanent tissues can be categorised into three types

(i) **Protective Tissue-Epidermis:**

- The epidermis is the outermost layer of the plant body and acts as the plant's first line of defence.
- It is a single layer of flat, rectangular cells packed tightly with no gaps.
- Epidermal cells are covered with a **waxy coating called the cuticle**, made of a substance called **cutin.**
- Differences between Parenchyma, Collenchyma and Sclerenchyma:

Feature	Parenchyma	Collenchyma	Sclerenchyma
Cell walls	Thin	Unevenly thick at corners (pectin)	Very thick (lignin)

Living or dead	Living	Living	Mostly dead
Cell spaces	Large spaces	Some spaces	No spaces
Main function	Storage, photosynthesis	Flexible support	Rigid support
Found in	Throughout plant body	Young stems, leaf stalks	Young stems, leaf stalks
Example	**Aerenchyma (aquatic plants)**	**Coriander stalks (soft)**	**Coconut husk, walnut shell**

Functions of the Epidermis:

- **Reduces water loss**: The cuticle prevents excessive loss of water. In dry-habitat plants, the cuticle is much thicker.
- **Protection:** The cuticle protects from mechanical damage and prevents entry of disease-causing parasites.
- **Root hairs:** In roots, epidermal cells extend into long hair-like projections called root hairs that greatly increase the surface area for absorbing water and minerals from the soil.
- **Stomata:** In leaves, the epidermis **contains tiny pores called stomata**, each bordered by **two guard cells** that control opening and closing.
    - Stomata allow exchange of gases ($C0_2$ in, 0_2 out) needed for photosynthesis and respiration.
    - They also help in **transpiration** (evaporation of water), which creates a **transpiration pull** in xylem that draws water upward from the roots.

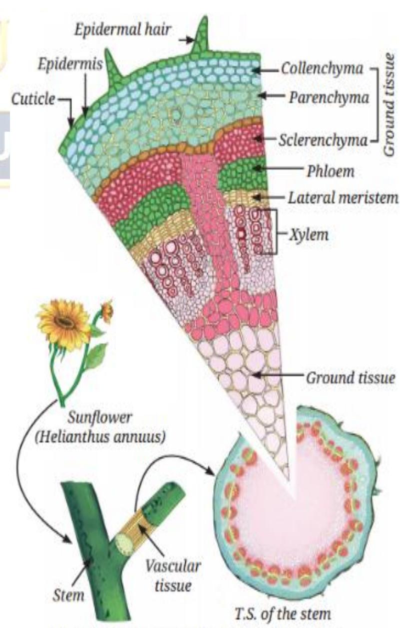
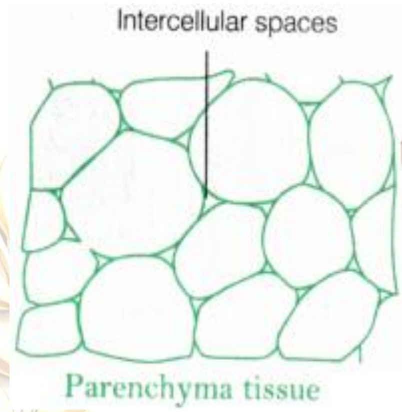


Fig. 3.7: Internal structure of a sunflower stem

(ii) Simple Permanent Tissues
These tissues **provide mechanical support** to keep the plant firm and upright. There are three types, each with a different structure suited to its specific role.

(a) Parenchyma

- It is the most common simple permanent tissue. These are **present in the soft parts** of the plant like leaves, roots, fruits, flowers, etc.
- It consists of simple living cells with little or no specialisation and thin cell walls.
- The cells are usually loosely packed with large spaces between them (intercellular spaces).
- It serves as a packaging tissue to fill the spaces between other tissues and maintains the shape of the plant.



They can be of two types

Chlorenchyma

- contains chlorophyll.
- can **perform photosynthesis**.

Aerenchyma

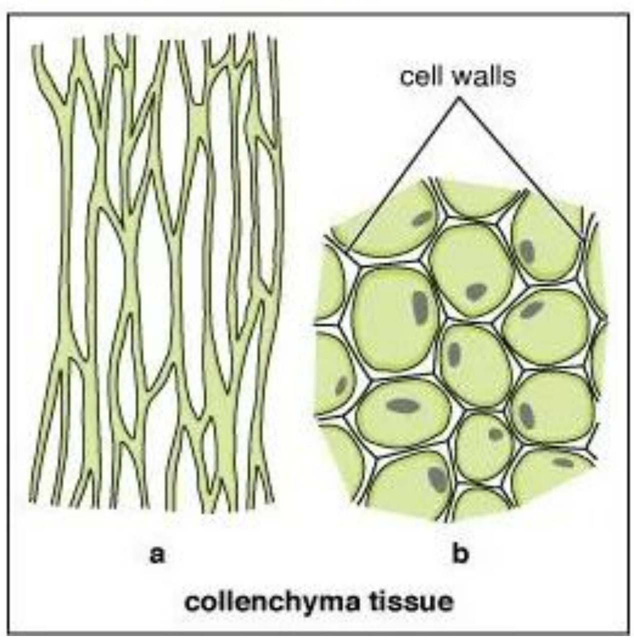
- contains large air cavities.
- **provides buoyancy** to aquatic plants.

Functions of Parenchyma

- Parenchyma serves as a food storage tissue.
- Parenchyma of stems and roots also stores nutrients and water.

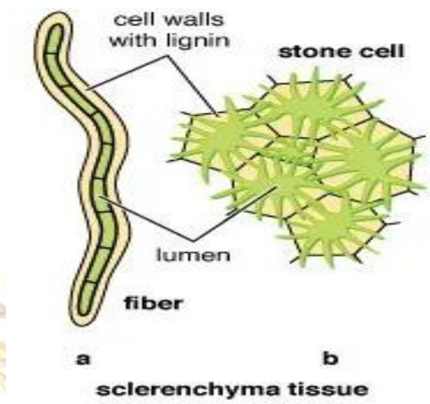
(b) Collenchyma

- These are living cells with unevenly thickened corners due to **deposits of pectin**, which gives the tissue a rubber-like flexibility.
- They provide strength to plant parts while still allowing them to bend without breaking.
- These are **found in young stems and leaf petioles** (stalks). For example, coriander leaf stalks are soft and flexible because of collenchyma.



(c) Sclerenchyma

- These are cells with very thick walls due to heavy deposition of lignin, making them hard and rigid.
- Most sclerenchyma cells are dead at maturity because the thick lignin walls block the movement of materials into and out of the cell.
- They provide hard, strong support and form the tough, woody parts of the plant.
- These are found in stems, leaf veins, and hard seed coats such as coconut husk and walnut shell.

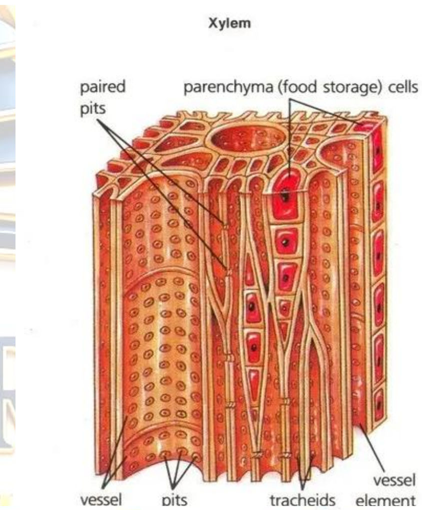


(iii) Complex Permanent Tissues

Complex permanent tissues are made of different types of cells that work together. Plants have two types of complex permanent tissues**; xylem and phloem**, together they are called vascular tissues.

(a) Xylem

Xylem transports water and dissolved minerals from the roots upward to all parts of the plant. It also provides mechanical strength.



Xylem is made of four types of cells

- **Tracheids:** Long, narrow, dead cells with thick walls. They transport water through small pores in their walls.
- **Vessels:** Wider dead cells arranged end to end forming long pipelines. They transport water faster and more efficiently than tracheids.
- **Xylem parenchyma:** The only living cells in xylem. They store food and assist in the sideways movement of water.
- **Xylem fibres:** Dead, thick-walled sclerenchymatous cells that provide mechanical strength and support.

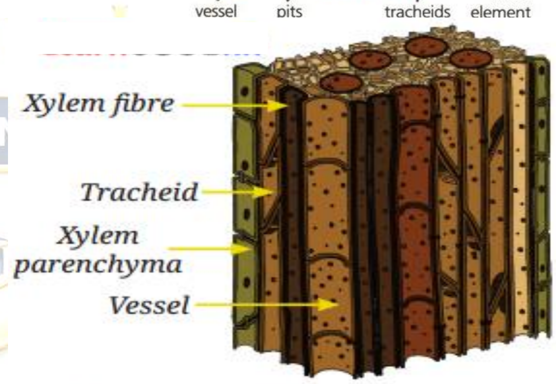


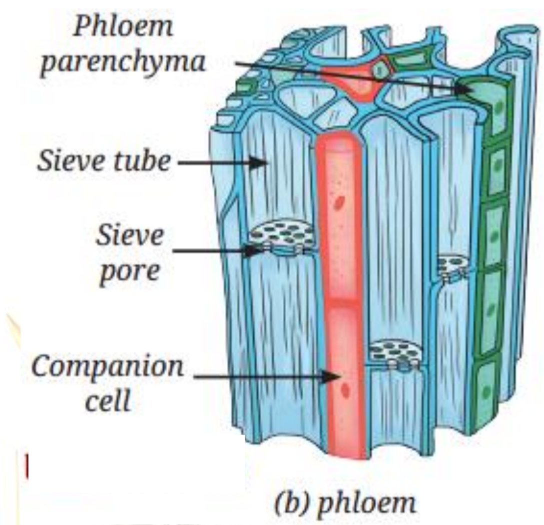
Fig. 3.9: Vascular tissue: (a) xylem,

(b) Phloem

Phloem transports **food (mainly sugars produced in leaves)** to all other parts(Upward and downward leaves to all parts)of the plant, including the roots. Unlike xylem, phloem is mostly made of living cells.

Phloem is made of four types of cells

- **<u>Sieve tubes</u>**: Long cells joined end to end by perforated walls (like a sieve). Food flows through these tubes from the leaves to other parts.
- **<u>Companion cells:</u>** Specialised parenchyma cells closely linked to sieve tubes. They regulate the loading and unloading of sugars into and out of the sieve tubes.
- **<u>Phloem parenchyma</u>**: Living cells that store food materials as well as substances like resin, tannins, and latex.
- **<u>Phloem fibres:</u>** Dead, thick-walled sclerenchymatous cells that provide structural support to the phloem tissue.



(b) phloem

<u>Totipotency:</u>

In 1958, F. C. Steward showed that even a single cell taken from the phloem of a carrot root could grow into a complete plant when placed in the right nutrient medium.

- The process involved two stages: **dedifferentiation** (the cell regained the ability to divide) followed **by redifferentiation** (cells became specialised to form roots, shoots, and the whole plant).
- This ability of a **single mature cell to develop into a complete organism** is called totipotency. Cells with this ability are called totipotent cells.

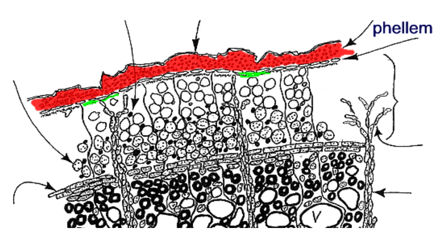
<u>Protective tissue</u>

<u>Epidermis</u>

- It is the outermost layer of all organs of the plant body which is formed from parenchymal cells. In epidermal cells outer walls are thicker than inner walls.
- It is mostly single layer but in desert plants it is multilayered for protection against water loss.
- **It protects the internal tissue from mechanical injuries and entry of germs**.

2. **<u>Cork or phellem</u>**

- Cork is the peripheral tissue of old stems and roots of woody trees and is formed due to activity of cork cambium or phellogen (secondary lateral meristem).



- Cork cambium (phellogen) produces new cells on its both sides, thus forming cork (phellem) on the outer side and the secondary cortex or phelloderm on the inner side.
- It is made up of dead cells with thick walls but no intercellular spaces.

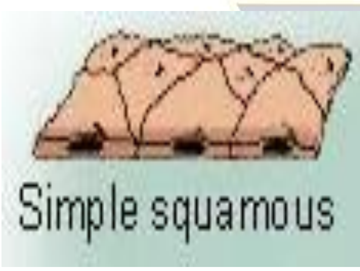
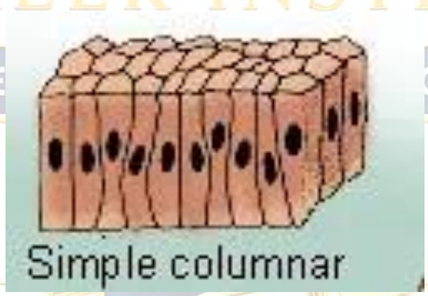
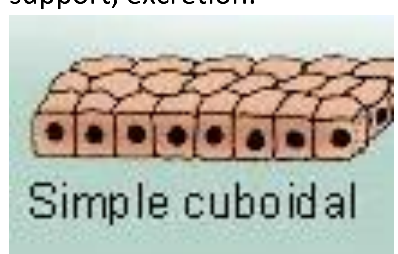
❖ **Animal Tissues**

Every animal, whether it is unicellular or multicellular, is capable of performing all vital functions such as respiration, ingestion, excretion and reproduction. Types of animal tissue: Based on the location and function, the animal tissues are **classified into four type**

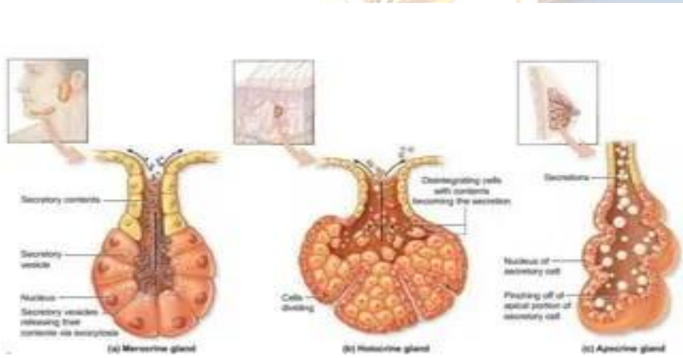
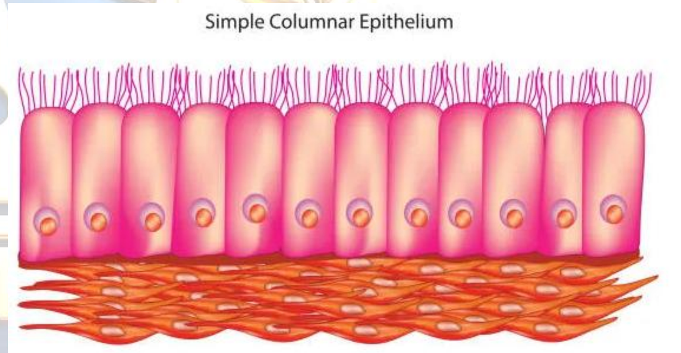
1. Epithelial Tissue

- It is the simplest tissue. It is the protective tissue of an animal's body.
- It **covers most organs and cavities** within the body. It also **forms a barrier** to keep different body systems separate.
- Epithelial cells are closely packed and have a small amount of cementing material, so there are very little intercellular spaces present between the cells.
- Due to absence or less of intercellular spaces; blood vessels, lymph vessels and capillaries are unable to pierce this tissue, **so blood circulation is absent in epithelium.**
- Hence cells depend for their nutrients on the underlying connective tissue.

Types of epithelial tissue on the basis of shapes and functions

Squamous epithelium (scale like)	**Columnar (Pillar like)**	**Cuboidal epithelium**
• Description : Flattened cells, extremely thin. • Common locations : Walls of blood vessels, air sacs of lungs, esophagus, lining of mouth. • Function : Diffusion  	• Description : Tall slender cells ; may have microvilli at its free surface. • Common locations : Inner lining of intestine, part of respiratory tract lining. • Function : Secretion, absorption  	• Description : Cube-like cells may have microvilli at its free surface. • Common locations : Lining of kidney tubules, ducts of salivary glands. It also forms the germinal epithelium of gonads. • Function : Secretion, absorption, mechanical support, excretion. 

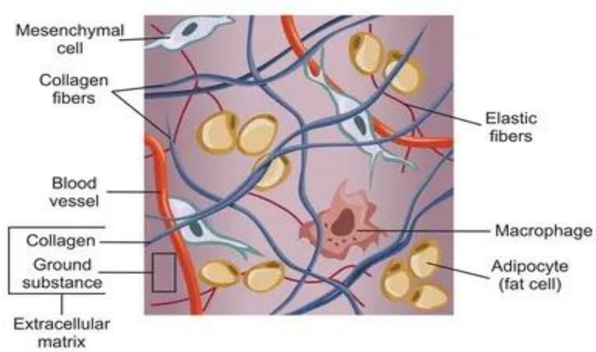
Modification of columnar epithelium

Glandular epithelium	**Ciliated columnar epithelium**
Description : Tall, slender cells, some cells from the free surface invaginate inside to form secretory cells – goblet cells. Common location : Lining of intestine & glands, trachea, bronchi. Function : Secretion of mucus and other secretions. 	Description : Tall , slender cells which possess cilia. Common location : Lines the nasal passages, oviducts, terminal bronchioles. Function : Protection, movement of substances in a particular direction for e.g. of mucus in nasal passages, egg in oviduct. 

2.Connective tissue

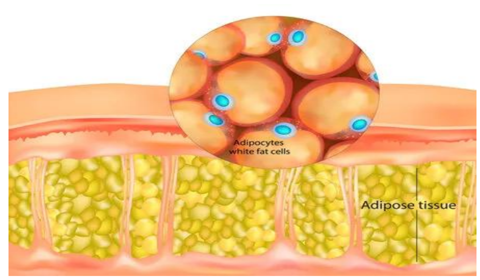
- The cells of connective tissue are loosely spaced and embedded into a non cellular matrix. The matrix may be solid (as in bone), soft (as in loose connective tissue), or liquid (as in blood).

Loose connective tissue



- Loose connective tissue is a mass of widely scattered cells whose matrix is a loose weave of fibres. Many of the fibres are strong protein fibres called collagen.
- **Loose connective tissue is found beneath the skin and between organs**.
- It is a binding and packing material whose main purpose is to provide support to hold other tissues and organs in place.

Adipose tissue



- It consists of adipose cells (Adipocytes) filled with fat globules in loose connective tissue.
- **Each adipose cell stores a large droplet of fat** that swells when fat is stored and shrinks when fat is used to provide energy.

- Adipose tissue pads and insulates the animal body

Fluid / vascular connective tissue

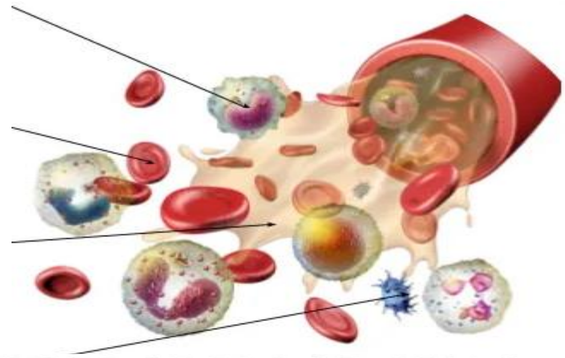
- It is a special type of connective tissue which maintains link among different parts of the body.
- It receives materials from certain parts of the body and transports them to the other parts. It constitutes the transport system of animals.
- It consists of two basic components – **blood and lymph**.
- Blood is a connective tissue of cells **separated by a liquid matrix called plasma.**
- It lacks fibers in its matrix.
- Blood constitutes 55 percent of plasma and 45 percent blood corpuscles.
- **Plasma contains water, salts, sugars, lipids, amino acids etc.**

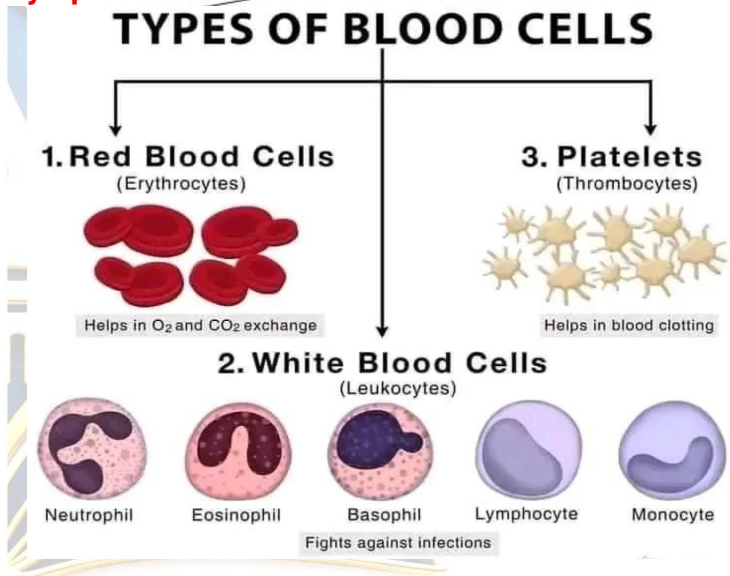
❖ Blood corpuscles are of 3 types :

(i) Red blood cells or erythrocytes.

(ii) White blood cells or leukocytes

(iii) Platelets or thrombocytes.





Lymph

•Lymph is actually filtered blood which is similar to blood in composition except that it is devoid of RBC, platelets and some blood protein.

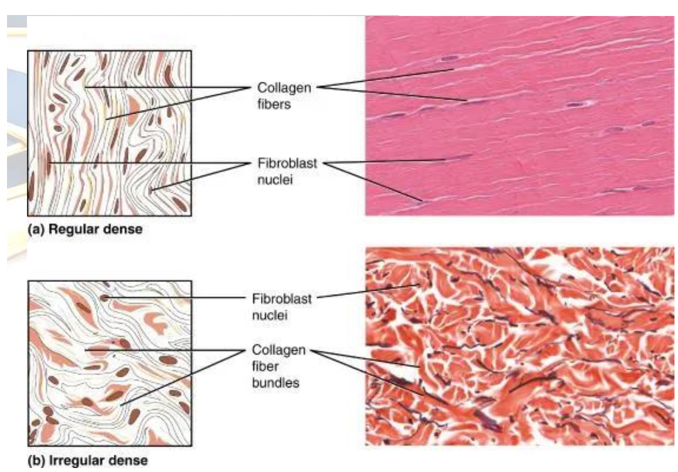
•**WBC are present in abundance in lymph**. Due to the absence of hemoglobin, lymph is **colorless**

Fibrous / dense regular connective tissue

This tissue consists mainly of fibers. The fibers are of 2 types :

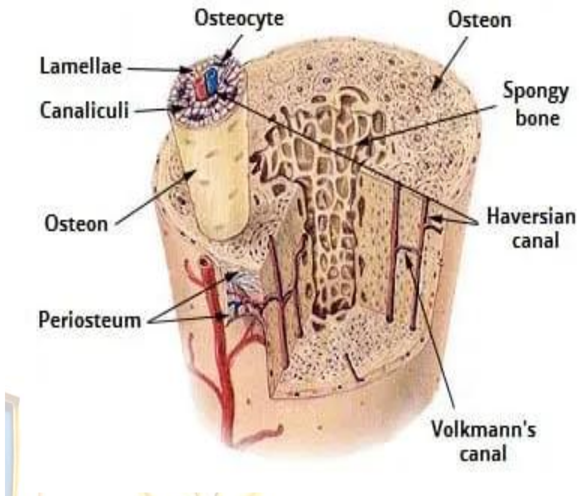
(1) **Collagen white fibers** The dominance of the (white inelastic) collagen fibers contributes to the considerable mechanical strength of white fibrous tissue.

(2) **Elastin yellow fibers** It can stretch upto one and a half times their length then snap back to its original length when relaxed



Skeletal connective tissue

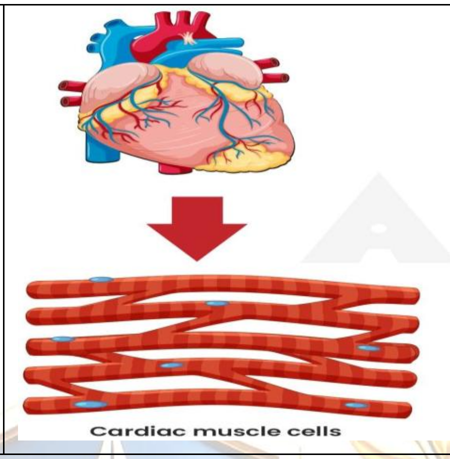
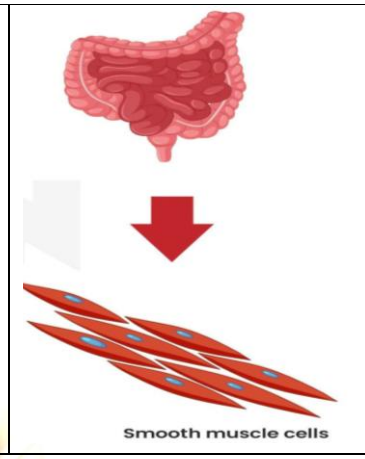
- It mainly consists **of bones and cartilages that provide a strong supportive framework** for the body.
- In these connective tissues, the matrix contains numerous fibers and in some cases, **deposits of insoluble calcium salts.**
- Bone It is a rigid connective tissue that has a matrix of collagen fibers and salt of calcium and phosphorous compounds, giving it greater rigidity and strength.
- Most of the skeletal system is bone. Haversian systems are the structural units of mammalian bone. It consists of Haversian canals, Haversian lamellae and osteocytes.



Muscular Tissue

- Muscular tissue is distinguished from other tissues by its **unique ability to contract & relax** and thereby perform mechanical work.
- It is responsible for movement of body organs and locomotion of the body.
- The structural unit of muscular tissue is the muscle cells which because of its elongated shape is also called muscle fibre.
- **The contractility is due to the presence of contractile proteins (actin & myosin) in the muscle fibre.**

Type of Muscle Tissue	Skeletal Muscle Tissue	Cardiac Muscle Tissue	Smooth Muscle Tissue
Location in body	**Attached to bones of the skeleton**. In the case of facial muscles, attached to other tissues including skin-hence also muscles expression". called as " of facial expression".	**Wall of the heart only**	**Walls of hollow internal structures,** includingBlood Vessels Stomach Intestines Gall bladder Urinary Bladder Airways to the lungs Iris of eye
Voluntary or involuntary	Voluntary	Involuntary	Involuntary
Cell Nuclei	Many nuclei	One	One
Cell Shape	Elongated cylindrical and unbranched	Cylindrical and branched	Spindle shaped and unbranched

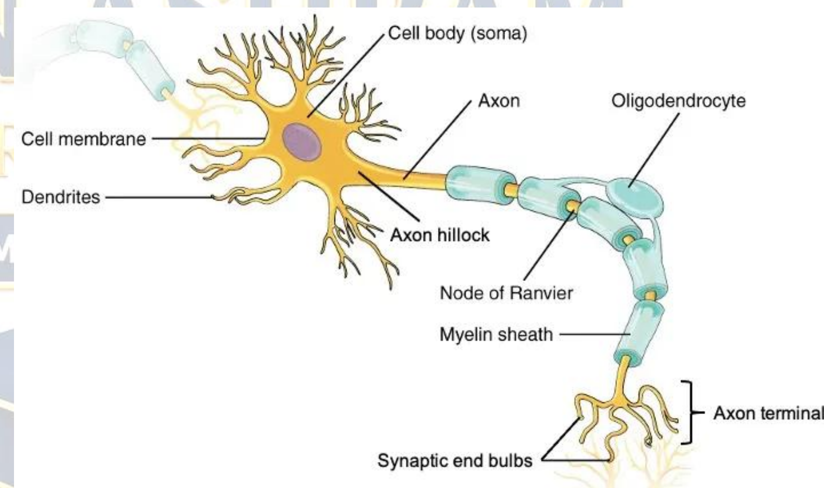
Nervous Tissue

- The nervous tissue, containing densely packed cells called nerve cells or neurons, is **present in the brain, spinal cord and nerves.**
- The **neurons are specialized for conduction of nerve impulses**.
- They receive stimuli from within or outside the body and conduct impulses (signals) which travel from one neuron to another neuron.
- **Nerve impulses allow us to move our muscles** when we want to.
- The functional combination of nerve and muscle tissue is fundamental to most animals.
- This combination enables animals to move rapidly in response to stimulus. Each neuron has the following 2 parts
    - **Cyton or cell body :** Contains a **central nucleus** and **cytoplasm** with characteristic deeply stained particles called **Nissl's granules** (i.e. clumps of ribosomes and rough endoplasmic reticulum).

(ii) Cell Processes

(a) Dendrites : These may be one to many, generally short and branched cytoplasmic processes. **Dendrites are afferent** processes because they receive impulse from a receptor or other neuron and bring it to the cyton.

(b) Axon : It is a single generally **long efferent process** which **conducts impulse away from cyton to other neurons**.



- **Longest cell in the body is the neuron because the axon can be more than one meter long**.
- Axon has uniform thickness but it has terminal thin branches called telodendria. Terminal end buttons or synaptic knobs occur at the end of telodendria
- Differences between Parenchyma, Collenchyma and Sclerenchyma: